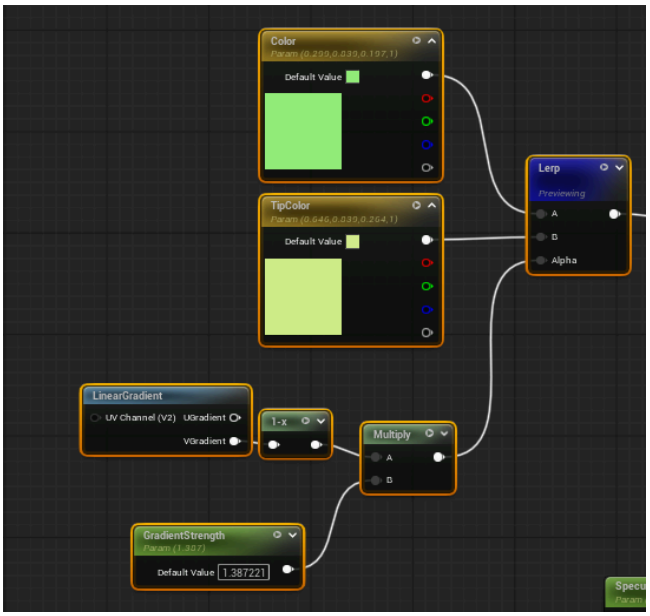
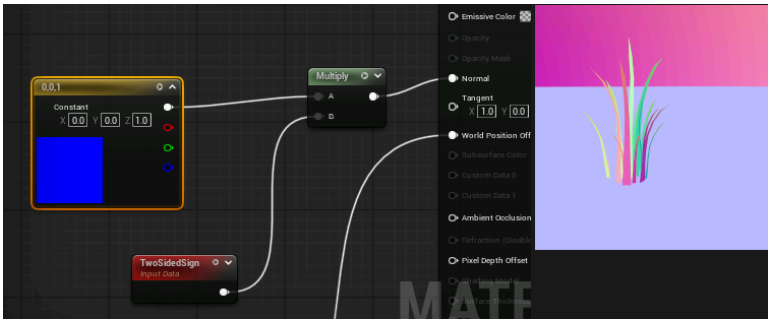
Documentation

GRASS SHADER



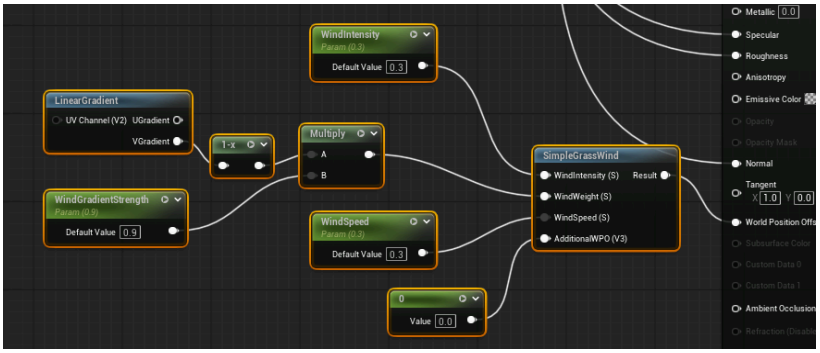
Explain image 1: making gradient colors to the object (grass).

1. Color
   Base-Color or first color to make a gradient colors
2. Tip Color
   The color at the tip (end) of a grass blade, the root color is near the ground.
3. Lerp (Linear interpolate) node
   Nodes used to blend between two values or colors based on Alpha mask.
4. Linear Gradient Node
   To control gradient value from U or V gradient. Because our grasslands are straight up so we can choose the V gradient or Y axis.
5. Minus-One Node
   In default color gradient comes out from the top to bottom so we can flip from bottom to the top.
6. GradientStrength (parameter)
   This parameter to control the gradient effect is.
7. Multiply
   Multiplies the inverted gradient by the GradientStrength value and connects to the alpha lerp node to control the gradient colors.



Explaining image 2: control normals to get the same directions front and back.

8. Two Sided Sign Node (Normals)
   Used to control normal in the same direction. Multiply with front-face by 1 and multiply back-face to -1.



Explaining image 3: wind effect to grass.

9. Linear Gradient Node
   It's used to make a gradient along the vertical axis (Y or V axis) of the grass blade. Used to control the wind's effect.
10. Minus-One Node
    Inverts the gradient to bottom become 0, top become 1.
11. WindGradientStrength (parameter)
    This parameter to control the wind effect.
12. Multiply Node
    This node takes the inverted gradient and multiplies it by the WindGradientStrength parameter. This allows you to control how much the gradient affects the wind.
13. WindIntensity (parameter)
    Controls how strong the wind is (overall sway).
14. Wind Speed (parameter)
    Controls how fast the wind makes grass sway.
15. 0 Constant
    It's connected to a part of the wind node that would normally be used for other types of movement. No additional world position offset, just using the wind.
16. SimpleGrassWind Node
    Simulates wind sway for grass/foliage.
    a. **WindIntensity** tells it how strong the wind is.
    b. **WindWeight** is the input for the gradient we created, telling the node which parts of the grass should move most.
    c. **WindSpeed** controls the speed of the sTway.
    d. **AdditionalWPO** is where the static **0** value is connected, ensuring no other movement is added.



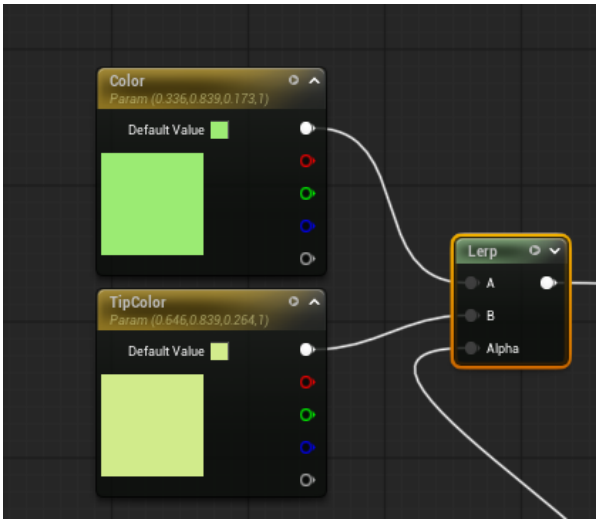
Explaining image 4:

17. Specular (parameter)
    Controls how strong the grass will reflect with a surface env. light. Grass can look more organic or plastics.
18. Roughness (parameter)
    Controls how sharp or blurry reflections are.

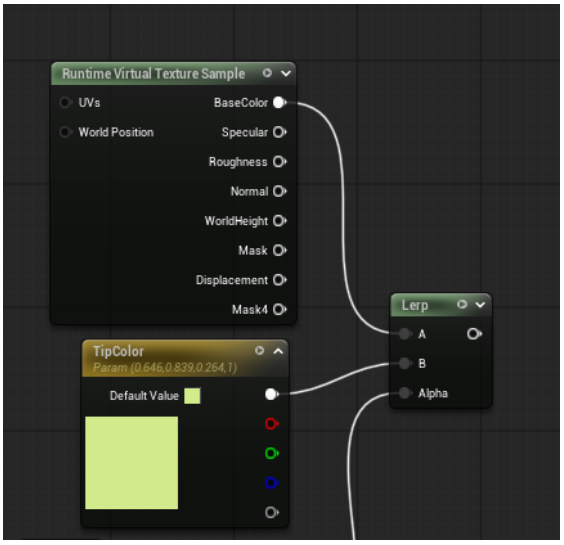
Short note on how the shader could be extended for production use:

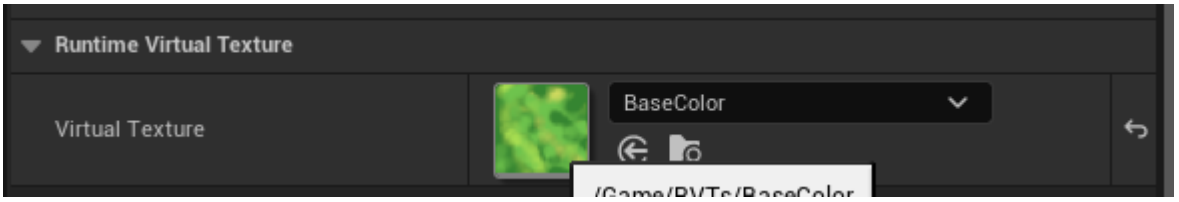
- Actually based on official documentation and best practice, I'd recommend adding LOD support. I'd also integrate it with the environment landscape RVTs to blend colors, compress textures, use less lighting and change it with a Two-Sided Foliage model for realistic lighting.
  Make sure overall it's modular so it can be reused across different assets.

1. Material grass shader explanation.

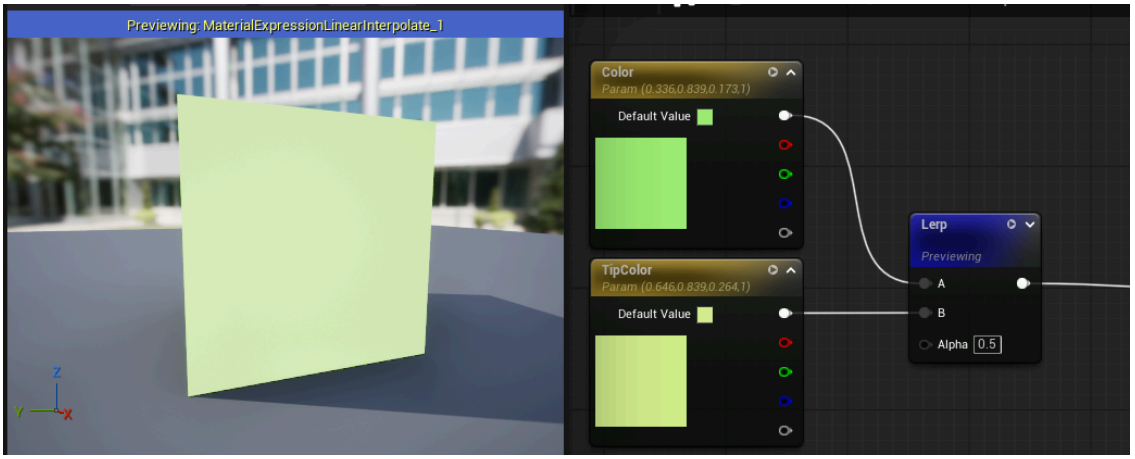


Im using 2 color variations to make color grading or color shading of my grass.

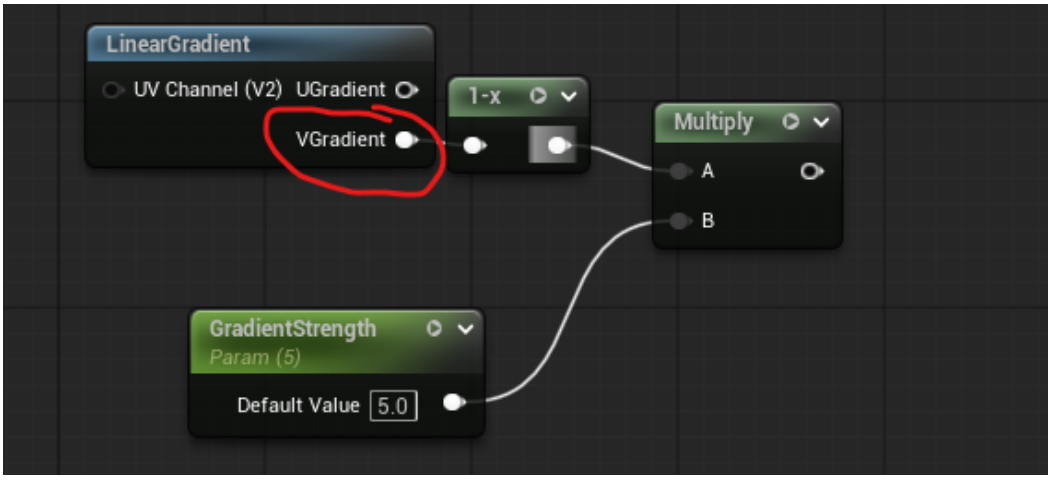




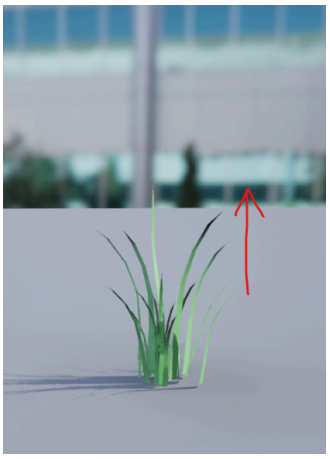
But we can also change base color with the RVT like above. The RVTs bake or stream your surface data (color, normal, height) into a virtual texture. It is best for optimizing large scale landscapes because there are fewer shader instructions at runtime.



and here the result of combination of the colors after connecting with lerp node.
Im using lerp node to make color grading and we can so handy control colors using alpha parameter in lerp node.

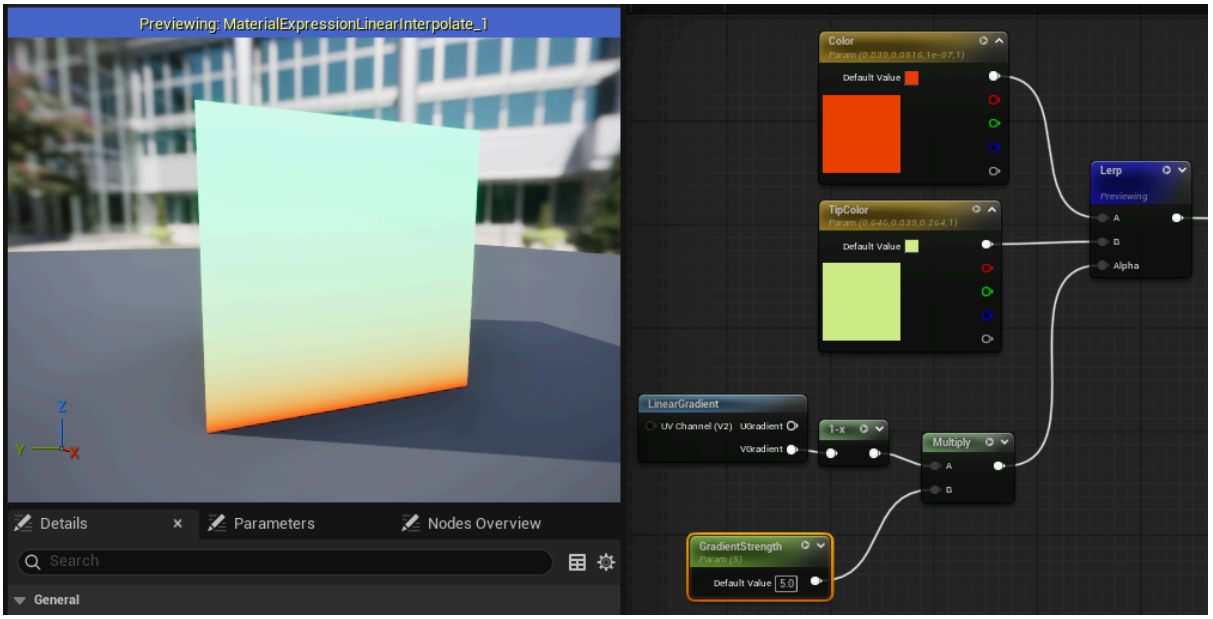


To set the gradient we using linear gradient node. so now we can see there are 2 different gradients u and v.



because our grasslands are standing straight up like this. so we choose the gradient in the y or v gradient axis. then connect to the v gradient axis with minus-one node to flip color because in default it is coming out from the bottom.

To control the intensity of the gradient effect, we create a 'gradient-strength' parameter.



and we can see what the gradient looks like.